Exp 3a – Electrons in Atoms

INTRODUCTION

When large amounts of energy are added to elements, such as when they are heated in a flame, they emit light. The light emitted is not every color of the visible spectrum, but specific colors (wavelengths). This happens because electrons in atoms are not just spread out evenly in the space around a nucleus but instead occupy specific energy levels or "regions" around the nucleus. To move to a higher energy level, an electron must absorb energy and when an electron moves to a lower energy level the energy must be released. The energy absorbed can come from many sources (heat, light, electricity, etc.), but energy released is mainly given off as electromagnetic radiation (light).

BACKGROUND

A spectroscope is a device used to separate light into its distinct wavelengths (a spectrum). In this experiment, we will use a simple spectroscope to analyze the visible light emitted from electron transitions in different atoms and obtain their corresponding line spectra.

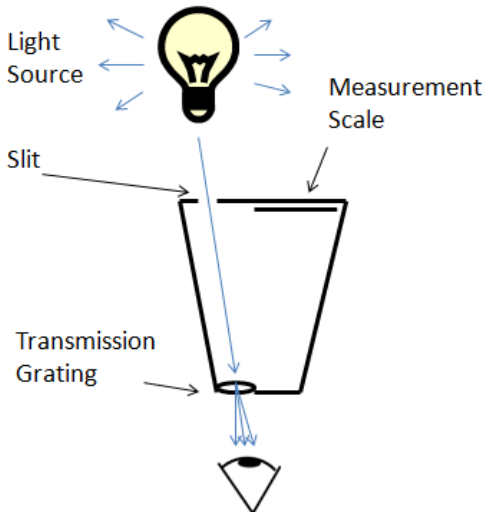


Figure 1: Spectroscope

The Rydberg equation (eqn 4) shows how the wavelengths (λ) of each line in the Hydrogen spectrum are related to small whole numbers, that wavelength is quantized.

$$\frac{1}{\lambda} = \frac{1}{91.13\ nm}\left(\frac{1}{{n_f}^2} - \frac{1}{{n_i}^2}\right) \qquad (4)$$

n_i = The "initial" shell the electron is falling from.

n_f = The "final" shell electron is falling to.

λ = The wavelength of light emitted.

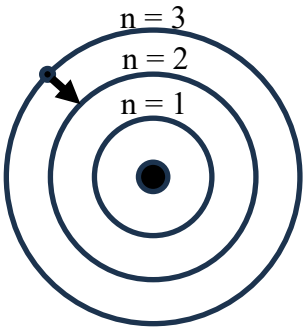


Figure 2: Bohr Model

The red line in the hydrogen spectrum corresponds to an energy change from the 3rd to 2nd level, the green line from the 4th to 2nd level, and the violet line from the 5th to 2nd level. *Energy is inversely proportional to wavelength (i.e., the energy change from the 5th to 2nd level is the largest) therefore the corresponding wavelength will be the shortest.* You will calculate the wavelengths of the red, green, and violet lines and compare them with your measured values.

ADDITIONAL READING

- The Bohr Model

SUPPLY LIST

- Spectroscopes
- Colored pencils

CLASSROOM SUPPLIES

These are shared supplies or reactants. Do not gather these items. Leave these items at their designated location in the classroom and use as needed according to the procedure.

- Element lamps

PROCEDURE

1. Aim the slit of the spectroscope at the brightest point in the lamp, then rotate your vision to see the line spectra on the measurement scale. (Once you aim the slit at the lamp, do not move the spectroscope-just move your head.)
2. Record the wavelength and color, of each strong line by drawing the spectrum on your report sheet as shown. (Use the colored pencil closest to the color of each emission line to draw the line on the chart.) If there are many lines in the same color, instead of recording the wavelength of each, record the wavelength range of the group. If you are having a hard time reading the measurement scale, shine a light (flash light, cell phone, etc.) on that part of the spectroscope while you are looking through the viewing window.

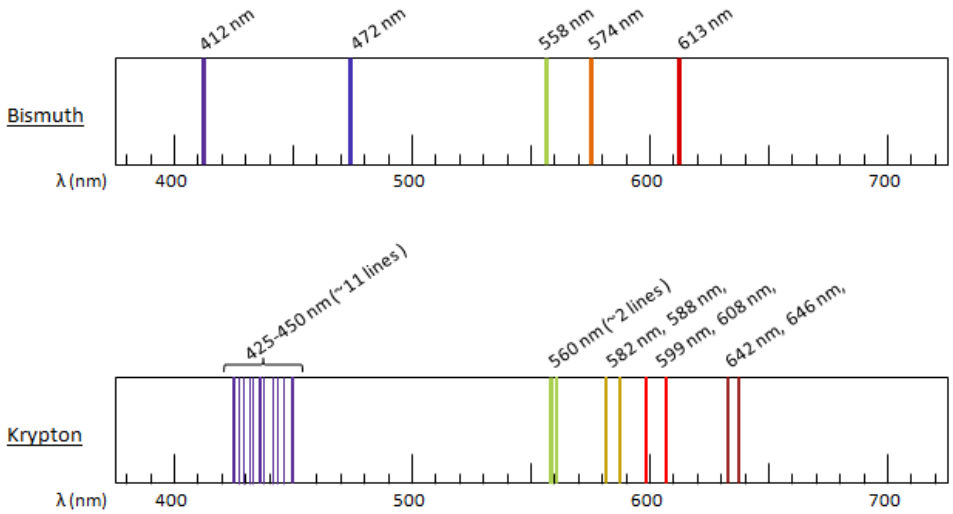


Figure 2: Emission Spectra for Bismuth and Krypton

* The scale on your spectroscope may be reversed, with the lower wavelengths on the right side of the scale. Be sure to note the actual position of the line(s) and record in the correct position on the provided charts.

* The drawings in figure 2 are what the line spectrum will look like on a high resolution spectroscope. When using the spectroscopes in this lab, the spectrum for Krypton will look something like this (figure 3).

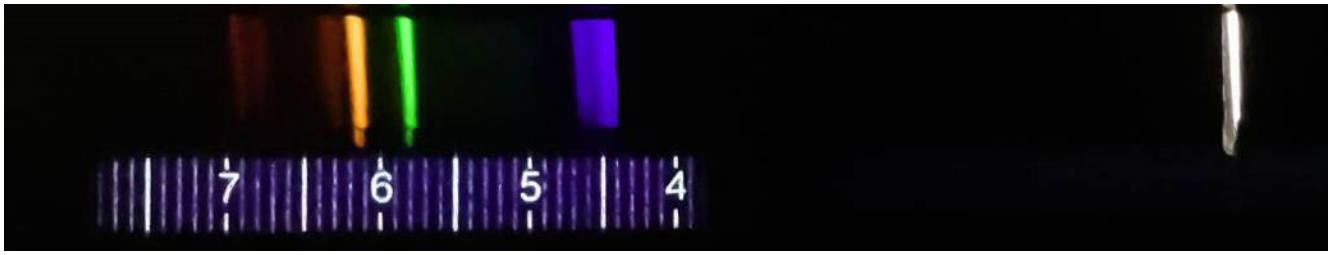


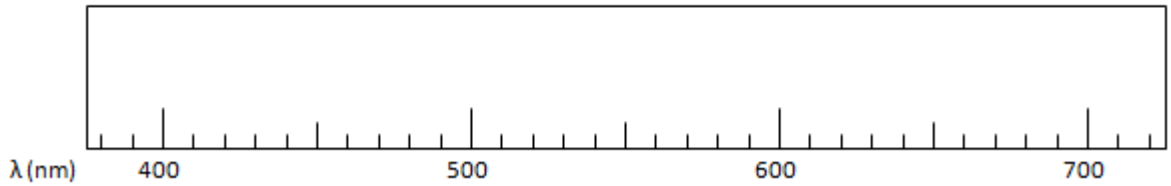
Figure 3: Emission Spectrum for Krypton on a Low-Resolution Spectroscope

DATA AND REPORT – DUE ____ / ____ / ____ **SECTION _____**

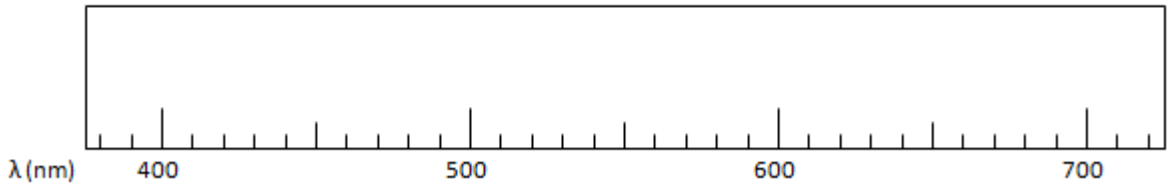
* Show your work and keep track of significant figures. **DATE _____**

PARTNER _____

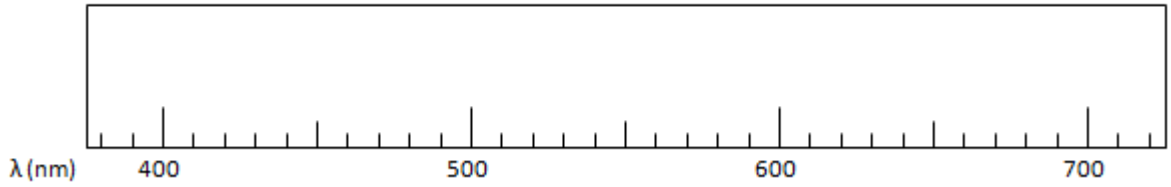
1. Helium (He)
   a. Use the spectroscope to analyze the light emitted by the helium lamp. Record your data below.



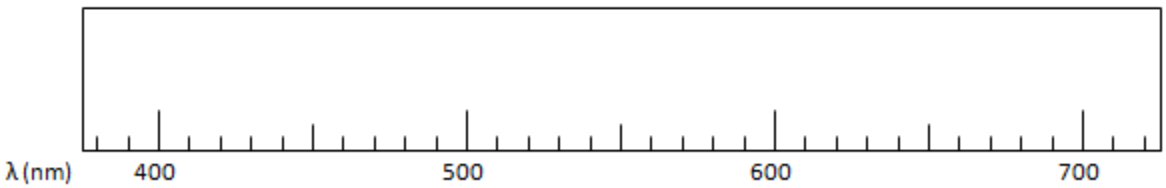
2. Nitrogen (N)
   a. Use the spectroscope to analyze the light emitted by the nitrogen lamp. Record your data below.



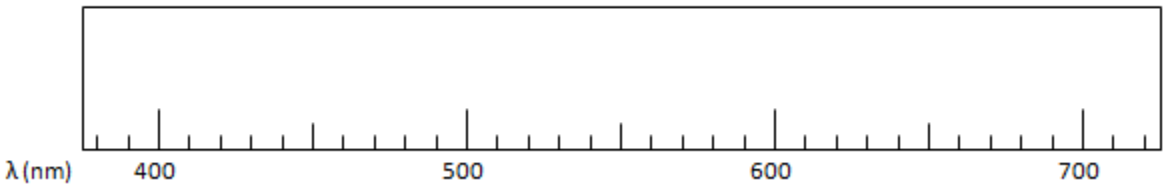
3. Neon (Ne)
   a. Use the spectroscope to analyze the light emitted by the neon lamp. Record your data below.



4. Mercury (Hg)
   a. Use the spectroscope to analyze the light emitted by the Mercury lamp. Record your data below.

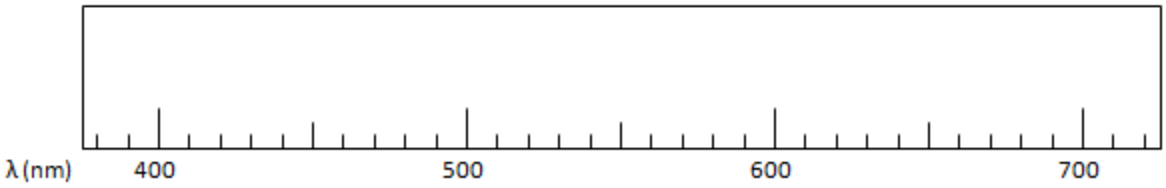


5. Air
   a. Use the spectroscope to analyze the light emitted by the Air lamp. Record your values below.



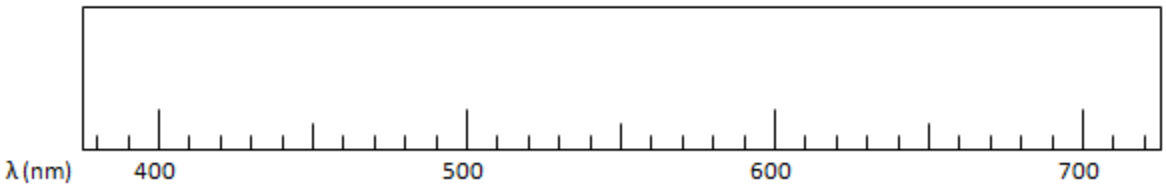
   b. Of the lamps you looked at earlier, which is most likely the main component of air? Why?

6. Mystery lamp
   a. Use the spectroscope to analyze the light emitted by the "Mystery" lamp. Record your values below.



   b. Of the lamps you looked at earlier, which lamp is most likely identity of the gas in the mystery lamp?

7. Hydrogen (H) – Measured wavelengths
   a. Use the spectroscope to analyze the light emitted by the hydrogen lamp. (Remember to record your data below as shown in the procedure section.)



Calculated Wavelengths for the Hydrogen spectrum

   a. Using the Rydberg equation calculate the wavelength of the violet line in the hydrogen spectrum. (From $n = 5$ to $n = 2$)

   b. Using the Rydberg equation calculate the wavelength of the green line in the hydrogen spectrum. (From $n = 4$ to $n = 2$)

   c. Using the Rydberg equation calculate the wavelength of the red line in the hydrogen spectrum. (From $n = 3$ to $n = 2$)

d. Use the table below to compare the values of your measured wavelengths with the values of the calculated wavelengths.

Line Color	**Measured Wavelengths from Spectroscope (nm)**	**Calculated Wavelengths from Rydberg Equation (nm)**	**Difference (Measured – Calculated)**
Violet			
Green			
Red			

e. Below, the three emission lines of Hydrogen are sorted in order of increasing wavelength. Fill in the wavelengths of each line, then complete the same ranking for frequency and energy.

Increasing wavelength

Violet line (____ nm) < Green line (____ nm) < Red line (____ nm)

Increasing frequency

Increasing energy

f. Arrange the 3 changes ($n = 3$ to $n = 2$, $n = 4$ to $n = 2$ and $n = 5$ to $n = 2$) in energy level in order of Increasing energy

g. Does the order of energy level changes in part g match the order of energy of the emission lines in part f?